Name – Arpita Waghmare

PRN-24070521006

SEC-A

```
%{
#include <stdio.h>
#include <stdlib.h>
int yylex();
void yyerror(char *s);
int my_pow(int a, int b) {
    int result = 1;
    for (int i = 0; i < b; i++) {
        result = result * a;
    }
    return result;
}
int my_div(int a, int d) {
    if (d == 0) {
        printf("Error: Division by zero\n");
        exit(1);
    }
    return a / d;
}
%}
%token NUMBER
%left '+' '-'
%left '*' '/'
%right '^'
%%
stmt: expr '\n'      { printf("Result: %d\n", $1);}
    ;
expr: NUMBER         { $$ = $1; }
    | expr '+' expr  { $$ = $1 + $3; }
    | expr '-' expr  { $$ = $1 - $3; }
    | expr '*' expr  { $$ = $1 * $3; }
    | expr '/' expr  { $$ = my_div($1, $3); }
    | expr '^' expr  { $$ = my_pow($1, $3); }
    ;
%%
void yyerror(char *s) {
    printf("Parser Error\n");
    exit(1);
}
int main(void) {
    printf("Enter expression: ");
    yyparse();
    return 0;
```

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```
%{
#include "y.tab.h"
#include <stdlib.h>
%}

%%

[0-9]+          { yylval = atoi(yytext); return NUMBER; }
"+"             { return '+'; }
"-"             { return '-'; }
"*"             { return '*'; }
"/"             { return '/'; }
"^"             { return '^'; }
"\n"            { return '\n'; }
[ \t]
.               { printf("Unknown character: %s\n", yytext); }

%%

int yywrap() {
    return 1;
}
```

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```
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/1006$ yacc -d prac9.y
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/1006$ flex prac9.l
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/1006$ gcc lex.yy.c y.tab.c -ll
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/1006$ ./a.out
Enter expression:
2 + 3 - 1
Result = 4
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/1006$ ./a.out
Enter expression:
2 / 0
Error: Division by zero
Result = 0
```